

Technology Stack

Date	26 June 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, Bootstrap	Used to create a simple, responsive, and user-friendly web interface for entering agricultural data and displaying crop recommendations.
2	Backend / Server-Side	Python, Flask	Flask connects the frontend with the machine learning model and processes user requests to generate crop recommendations.
3	Database / Data Storage	CSV Dataset (Crop_recommendation.csv), Pickle (model.pkl)	The CSV file stores agricultural data for training, and the Pickle file stores the trained model for fast prediction.
4	Cloud / Hosting / Deployment	Flask Local Development Server	Used to run and test the application locally during development before deployment.

5	Version Control & CI/CD	Git, GitHub	Used to manage source code, track project changes, and maintain different versions of the application.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, VS Code	These libraries are used for data preprocessing, machine learning model training, visualization, analysis, and application development.